Production of English /r/ by Prosodic Position

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Preview

- ❖ Does production of English /r/ depend on prosodic position?
- ❖ Does it depend on local vowel and consonant context?
- ❖ We carried out an ultrasound study of 5 English speakers.
- ❖ We find:
  - ➢ /r/ is surprisingly unaffected by prosodic position.
  - ➢ Retroflexion is relatively incompatible with a context of [t] or [i] (replicating previous work).

Preliminaries

- A good deal is known about the production of English /r/ according to segmental context or position in the syllable [2,3,6], e.g.:
  - ○ "Retroflex" /r/ more likely in syllable onset, and around back/low vowels, and around labial consonants.
  - ○ "Bunched" /r/ is more likely in coda, around high/front vowels, and around coronal consonants.
- Less is known about whether the production of /r/ depends on *higher* prosodic context.
- Phrase- and word-initial position are known generally for longer, more hyper-articulated consonant productions [4].
  - ○ What might that mean for /r/?
- Here we compare the production of /r/ in syllable-initial, word-initial, and phrase-initial positions (always stressed).

Ultrasound study

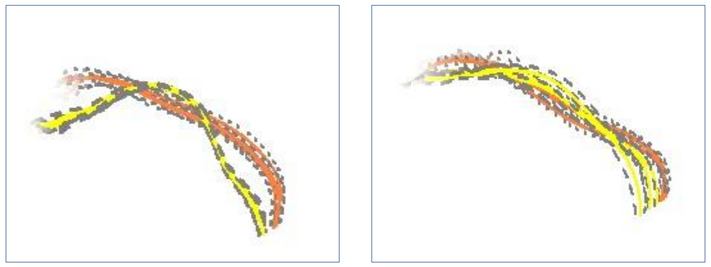
- 5 participants, university students recorded at UCSC.
- Produced /r/ in onset of stressed syllable in carrier phrases. 18 conditions:
  - Phrase- vs. word- vs. syllable-initial
  - /tr/ vs. /pr/ vs. /r/ (all in onset)
  - Followed by [i] or [ɑ]



- 6 repetitions in each condition.
- Micro ultrasound system with head stabilization.
- Analyzed using Articulate Assistant Advanced software [1].
- Results here based on average tongue curves. We followed [5] in determining whether a tongue shape was "bunched" vs. "retroflex".
- Example carrier phrases:
  - Phrase Initial: *Don't do the play "Sofa". "**Reed**" is better.*
  - Word Initial: *Don't do the play "Sofa". "Sandal and **Reed**" is better.*
  - Syllable Initial: *Don't do the play "Sofa". "Sandal and Be**reave**" is better.*

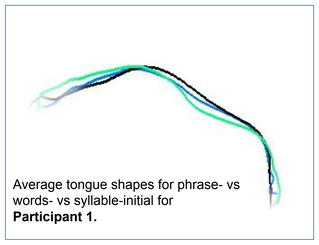
Results

- Our results are clearest when we compare /r/ in the context t_i vs. p_a. The context t_i favors a more "bunched" tongue shape while p_a favors a more "retroflex" tongue shape.
- /r/ is more stable across prosodic positions than we thought it would be. In the figures below, each orange and yellow "curve" is actually 3 separate curves for syllable-, word- and phrase initial productions, overlaid. They're essentially identical!

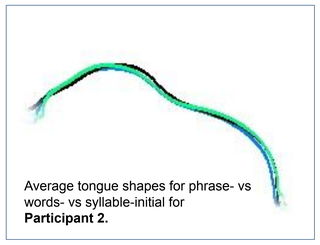


Average tongue shape of phrase- vs word- vs syllable-initial productions for Participant 4 (right) and 7 (left). Orange is p_a context and yellow is t_i context.
Standard Deviations displayed as dotted (orange) and dashed (yellow) lines.

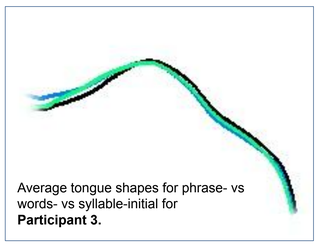
- Replicating previous findings (e.g. [2,3,6]), our participants tend to produce a more "retroflex" /r/ in the p_a context and a more "bunched" /r/ in the t_i context.
- For most participants the difference is subtle, as on the upper right. Note the tongue blade lowering with slight concavity in the tongue body in the t_i context.
- Participant 7 (upper left) makes more of a categorical bunched (t_i) vs. retroflex (p_a) distinction.
- The figures below show /r/ with no preceding consonant (e.g. *read, rod*) across the 3 prosodic positions (phrase-, word-, syllable-initial), combining the _i and _a contexts. Again we see how stable /r/ is across positions.



Average tongue shapes for phrase- vs words- vs syllable-initial for **Participant 1.**



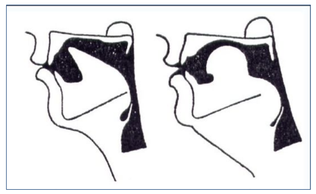
Average tongue shapes for phrase- vs words- vs syllable-initial for **Participant 2.**



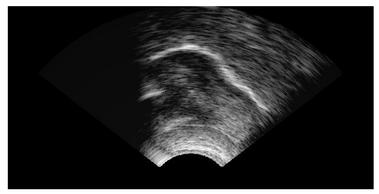
Average tongue shapes for phrase- vs words- vs syllable-initial for **Participant 3.**

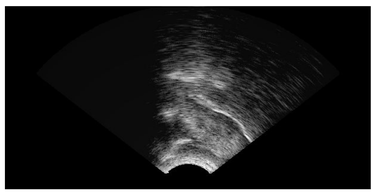
Discussion

- ❖ 4 of our 5 participants produced a more "bunched" /r/ overall, as on the right in this figure [2].
- ❖ But all showed a tendency towards more retroflexion in the p_a context. This is probably because retroflexion is somewhat incompatible with the gestures needed for [t] and [i] [2,3,6].



- ❖ Relatedly, the participants generally favored *one* variation over the other. Participants 1 to 3 had a bias toward bunching, and participant 4 had a bias to toward retroflexion.
- ❖ These participants only had slight variation in aspects like tongue root advancement and angle of apical direction. These features occurred while maintaining an overall bunched or retroflexed form.





- ❖ /r/ is highly stable across prosodic positions.
- ❖ This is surprising given the literature on domain-initial strengthening [e.g. 4], which shows that consonants are longer and more hyper-articulated in higher prosodic positions.
- ❖ However, /r/ has been shown to resist coarticulation and be less subject to allophonic variation, at least compared to /l/ [7]. Whatever is behind this, maybe it explains the stability of /r/ across prosodic positions too.

References

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